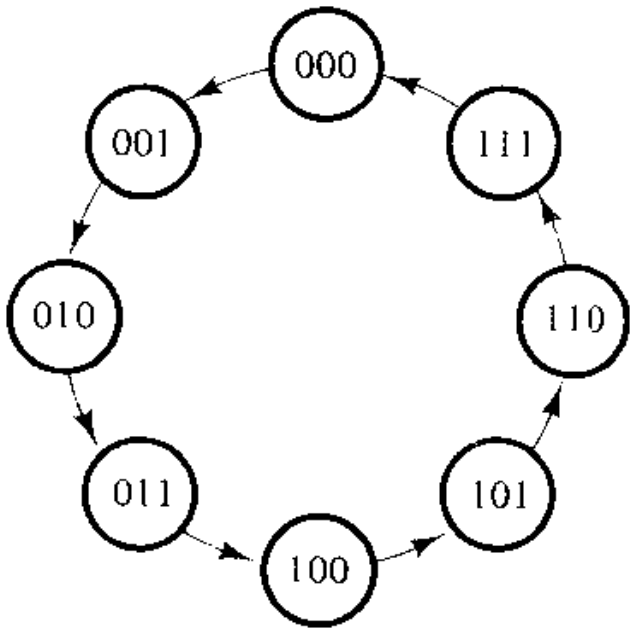
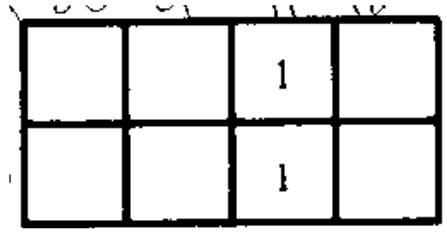
Design of synchronous counters

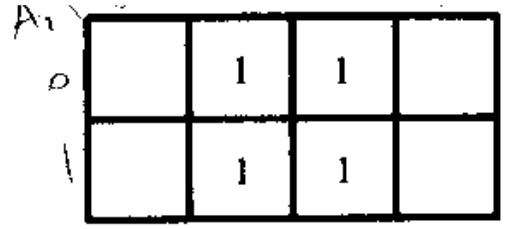


Excitation Table for 3-Bit Counter

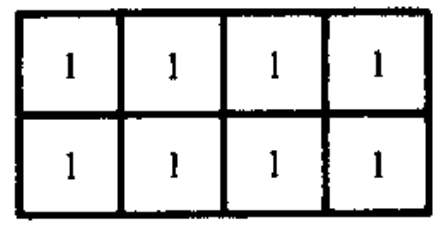
Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	TA_2	TA_1	TA_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1



$TA_2 = A_1 A_0$



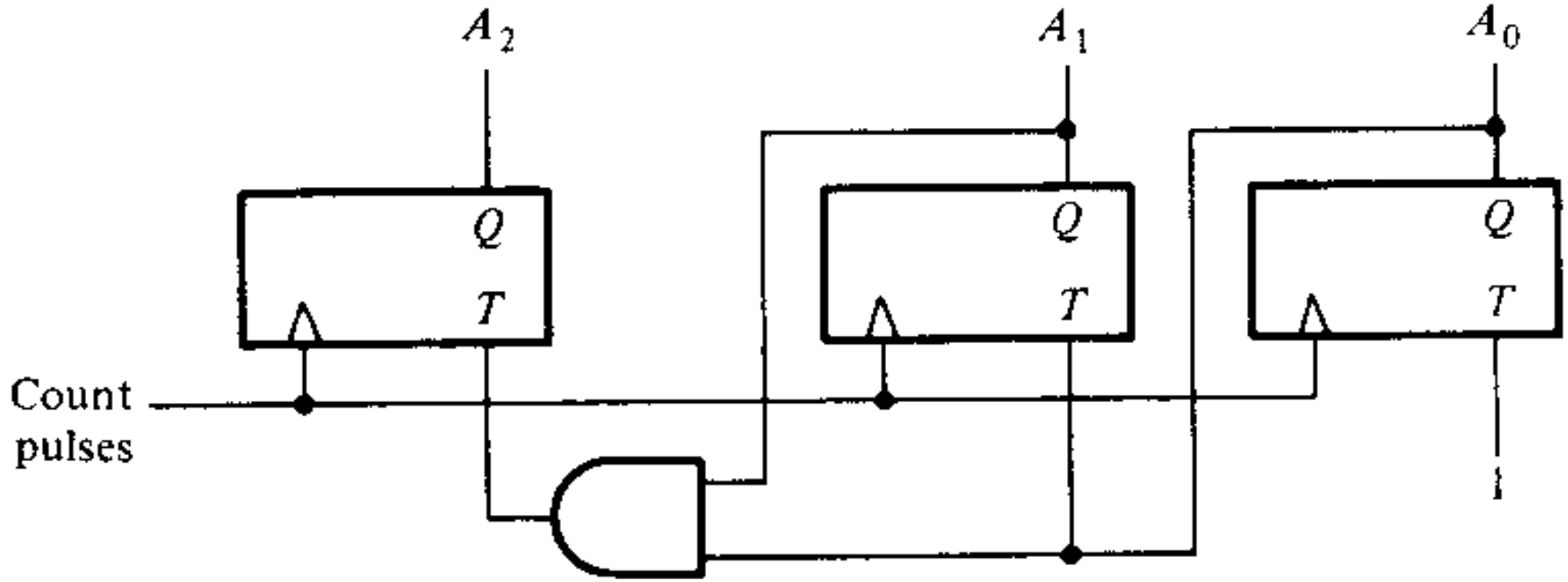
$TA_1 = A_0$



$TA_0 = 1$

Excitation Table for 3-Bit Counter

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	TA_2	TA_1	TA_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1


A_2
A_1
A_0
Q
T
Q
T
Q
T
Count
pulses

Counter with Nonbinary Sequence

Design a three-bit counter that counts in the sequence 0, 1, 2, 4, 5, 6, 0, ……

$Q(t)$	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(a) JK Flip-Flop

Excitation Table for Counter

Present State			Next State			Flip-Flop Inputs					
A	B	C	A	B	C	JA	KA	JB	KB	JC	KC

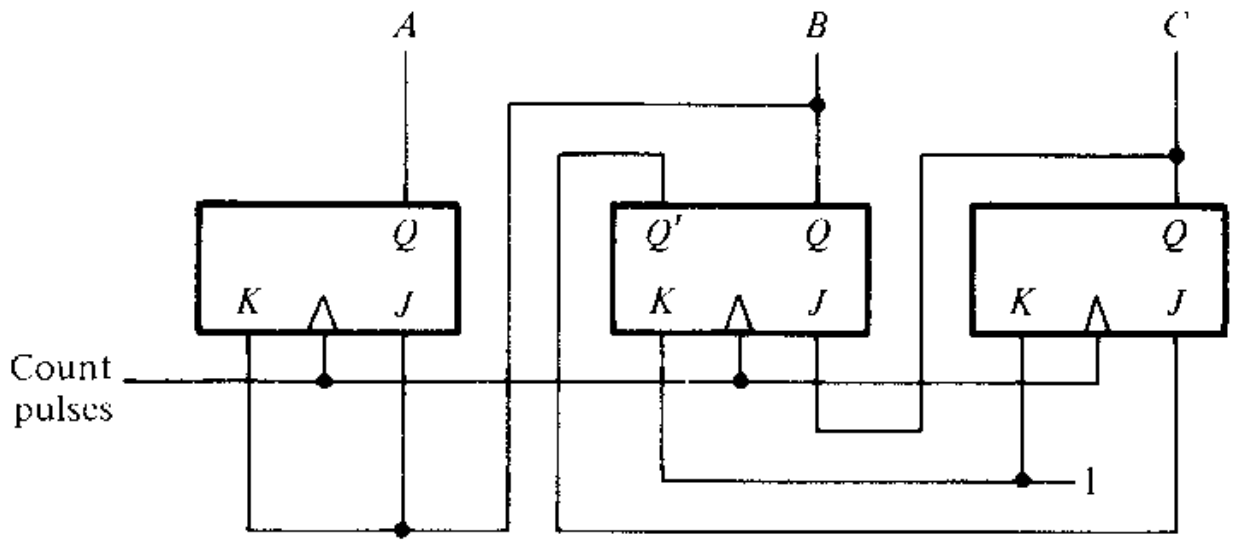
Excitation Table for Counter

Present State			Next State			Flip-Flop Inputs					
A	*B*	*C*	*A*	*B*	*C*	*JA*	*KA*	*JB*	*KB*	*JC*	*KC*
0	0	0	0	0	1	0	*X*	0	*X*	1	*X*
0	0	1	0	1	0	0	*X*	1	*X*	*X*	1
0	1	0	1	0	0	1	*X*	*X*	1	0	*X*
1	0	0	1	0	1	*X*	0	0	*X*	1	*X*
1	0	1	1	1	0	*X*	0	1	*X*	*X*	1
1	1	0	0	0	0	*X*	1	*X*	1	0	*X*

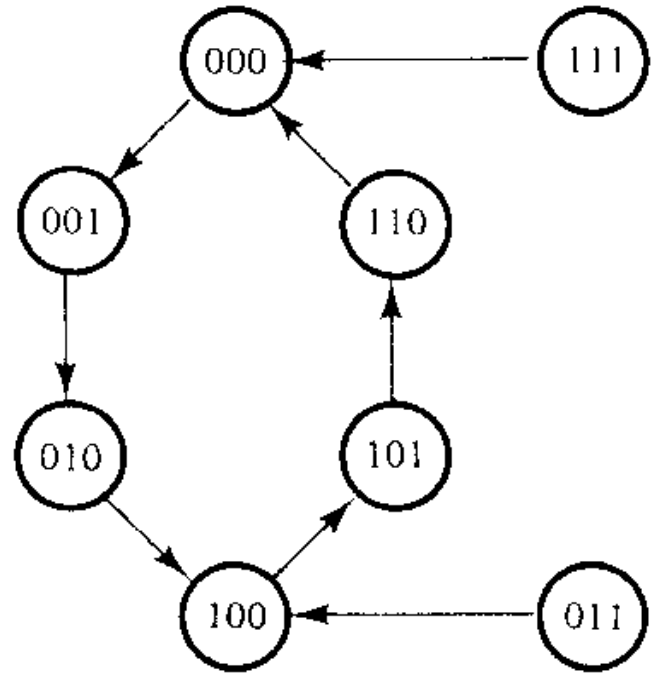
$$JA = B \qquad KA = B$$

$$JB = C \qquad KB = 1$$

$$JC = B' \qquad KC = 1$$



(a) Logic diagram of counter



(b) State diagram of counter

Problem

A *JN* flip-flop has two inputs, *J* and *N*. Input *J* behaves like the *J* input of a *JK* flip-flop and input *N* behaves like the complement of the *K* input of a *JK* flip-flop (that is, $N = K'$).

(a) Tabulate the characteristic table of the flip-flop (as in Table 6-3).

(b) Tabulate the excitation table of the flip-flop (as in Table 6-10).

(c) Show that by connecting the two inputs together, one obtains a *D* Flip-flop.

Problem:

Design with the following repeated binary sequence 0, 1, 3, 7, 6, 4.

Use T FF. Check if it is self correcting

Home work

(a) Design a counter with the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6. Use *JK* flip-flops.
(b) Design a counter with the following repeated binary sequence: 0, 1, 2, 4, 6. Use *D* flip-flops.
(c) Design a counter with the following repeated binary sequence: 0, 1, 3, 5, 7. Use *T* flip-flops.

Check all of your design to see if it is self correcting or not